

Difficulty–Response Curves: A Form Guide for Reasoning Models

alex newman

posix4e@gmail.com · code, data and generators: github.com/posix4e/difficulty-response-curves

Working notes — a living document reporting what we have learnt so far. Measurements are ongoing; corrections are published in place and kept in the text as part of the findings. Last updated 6 July 2026.

Abstract. A leaderboard gives every AI model one number, and one number cannot tell you the thing you actually need to know before you rely on a model: as problems get gradually harder, where does its success rate collapse, and how suddenly? We measure exactly that. We generate logic puzzles with a **difficulty dial** — random 3-SAT, where turning the dial adds constraints to a puzzle whose answer we can check mechanically — and test eight reasoning models many times at each of fifteen dial settings. Plotting pass rate against difficulty gives each model an S-shaped **difficulty–response curve**, which two numbers summarise: the **frontier** x_{50} (the difficulty at which the model succeeds half the time) and the **sharpness** a (how abruptly success collapses past it). Psychometricians have fitted exactly these curves to humans for seventy years under the name item response theory; we point the same instrument at machines and ask whether the readings can be trusted. They can. First, re-measuring on freshly generated puzzles reproduces the numbers (split-half reliability $r = 0.99$ across models; the two models re-tested in full repeated to within 0.02 and 0.12 dial units) — more than most benchmark scores can say. Second, the frontier is mist, not a wall: at the frontier, 81 per cent of outcome variance is luck of the draw on the **same** puzzle rather than differences between puzzles, so retrying a failed problem there genuinely helps. Third, a model’s spending of reasoning tokens peaks within 1.08 dial units of its frontier, so the cliff can be located from billing metadata alone — no internal access needed. Fourth, a simple binary search finds the frontier with $6.8\times$ fewer tests than a full sweep. Generators, raw data, fitting code, and a tool to measure your own model are released. No mechanism, no grand theory of mind. An instrument.

1 Introduction

Suppose you are about to trust a language model with real work, and the work arrives at varying levels of difficulty. The question that matters is not “how smart is it” — leaderboards answer that with a single score, and single scores hide everything useful. The question is: **as the work gets harder, where does this model stop coping, and does it fail gently or fall off a cliff?** Everyone who has used a reasoning model has watched it glide through medium problems and then collapse two ticks later. The pattern is well photographed in the research literature on reasoning “collapse”. What nobody hands you is the measuring device: for this model, on this kind of problem, where the cliff sits, how wide the crumbling ledge is, and whether the reading would hold if you measured again tomorrow.

This paper builds that device. The recipe has four steps and no magic. **One:** generate puzzles by computer, with a knob that controls difficulty — our main family is random 3-SAT, a classic logic puzzle where the solver must find a true/false setting for twenty variables that satisfies a list of constraints, and turning the knob simply adds more constraints. Because the puzzles are generated fresh, the model cannot have memorised them, and because a proposed solution can be checked mechanically, grading involves no judgement calls. **Two:** test a model repeatedly at each of fifteen knob settings, from easy to hopeless. **Three:** plot the fraction it solves against the knob. The result is always an S-shape: near-certain success on the left, near-certain failure on the right, a slope between. **Four:** fit that S-shape and read off two numbers — the **frontier** x_{50} , the knob setting where success crosses fifty-fifty, and the **sharpness** a , how steep the fall is. Psychometricians have fitted exactly these curves to human test-takers since the 1950s, under the name item response theory; the mathematics arrives ready-made, and this paper’s only job is to show that it transfers to machines and to report what it finds.

Why “a form guide”? At the races, the pamphlet in every punter’s hand never says whether a horse is **good** — a meaningless question — but what distance it runs, on what ground, and whether it fades in the final straight. That is the shape of answer a model deserves: a datasheet of measured, repeatable numbers

about where it runs out of legs (Table 1 is literally that table). The Hong Kong Observatory does not explain the typhoon either. It hoists a numbered signal, and everybody knows what T8 means.

We make four claims. Each is registered here with the manner of its death printed beside it — the result that would have falsified it — because an instrument you cannot break is not an instrument.

- C1, reliability** measure a model twice — fresh puzzles, fresh attempts — and you get the same frontier and sharpness back. Dies if the error bars are so fat that the model rankings reshuffle between Tuesday and Thursday.
- C2, the misty frontier** at the frontier, when a model fails a puzzle, is that because the puzzle was secretly harder, or because the model is simply a coin-flip there — the **same** puzzle passing on one attempt and failing on the next? We measure the split. Cannot lose, only cannot lie: if it is mostly the puzzles, then the “frontier” is just a sorting of hard problems from easy ones, and we say so in exactly those words.
- C3, the effort marker** models spend the most reasoning tokens — visible in any API bill — on problems near their own frontier, so token counts alone can locate the cliff without any internal access. Dies if the spending peak is missing or wanders about uncorrelated with the frontier across models.
- C4, fast estimation** you should not need thousands of tests to find a frontier; a binary search — try a middling difficulty, jump harder on success and easier on failure — recovers it at a fraction of the cost of testing every level uniformly. Pure engineering. We state the ratio and do not dress it up in a dinner jacket.

2 The instrument

2.1 Task families

Random 3-SAT, satisfiable only, certificate required. The primary family is random 3-SAT at fixed $n = 20$ variables, with the clause-to-variable ratio α swept across fifteen levels from 2.0 to 5.6; an instance at level α has $m = \text{round}(\alpha n)$ distinct clauses of three distinct variables with independent random polarities. Theorists gift-wrapped this difficulty axis decades ago [1], [2]. We add one design decision: every instance is rejection-sampled at generation time to be **satisfiable**, and the model must output a complete assignment, which we check directly against the clauses. A pass is a verified certificate. This kills the guessing floor — a random assignment satisfies all m clauses with probability $(7/8)^m$, under half a per cent at the easiest level — so the response curve needs no third guessing parameter, and verification needs no trust in anything. The cost of the decision is stated plainly: conditioned on satisfiability, high- α instances are not typical of random 3-SAT; the knob measures the constrainedness of solvable instances, which is what an instrument for **reasoning up to a frontier** wants anyway.

DAG arithmetic, the second opinion. Random arithmetic circuits over small integers: leaves in $[-9, 9]$, operations from $\{+, -, \times\}$, each new node consuming the immediately previous node plus one earlier node, so dependency depth equals the op count by construction. Any intermediate exceeding $|v| > 999$ is locally resampled: per-step arithmetic stays small, and difficulty comes from length and structure, not from multiplying big numbers. The knob is the op count across ten levels from 2 to 42. Pass is exact integer match.

Contamination. Every instance is freshly generated from named seed streams; retest sets are disjoint by construction. A fixed canary string rides in every prompt as a tripwire for future training runs.

2.2 The curve

Higher x means harder. We fit

$$P(x) = (1 - \lambda) \cdot \sigma(a(b - x)), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

a two-parameter logistic with an upper lapse $\lambda \in [0, 0.15]$ and the floor fixed at zero (the certificate design bought us that). The lapse matters: models drop the odd easy instance to format slips and carelessness, and a curve forced through a 100 per cent ceiling buys that slack by biasing the frontier. The reported frontier is $x_{50} = b - \logit(0.5/(1 - \lambda))/a$, the level where the **fitted** pass rate crosses one half.

Point estimates are penalised maximum likelihood on per-instance binomial counts: a mild Normal prior on $\log a$ keeps the slope finite when the data are step-like, and a gentle Beta prior shrinks the lapse toward small values. Every confidence interval in this paper is a stratified cluster bootstrap over **instances** (an instance carries all its samples; levels are design points and are never resampled), 2000 replicates, percentile intervals. With heterogeneous instances this likelihood is deliberately the **population-averaged** curve: a is a marginal slope, flatter than any single instance’s, and that is the right object for a form guide. Goodness of fit is a level-wise Pearson χ^2 calibrated by parametric bootstrap under beta-binomial instance effects matched to the observed frontier ICC, because the plain binomial reference over-rejects under clustering.

3 Protocol

Models. Six core models across four families and two within-family scale pairs, plus two budget-contingent extras, all served through one OpenAI-compatible gateway (TrustedRouter) with the provider pinned per model and the serving endpoint recorded on every call. Models that do not reason unless asked were excluded by construction — the roster is models that reason by default. Exact model strings, providers, prices, and decoding settings are in Section 7.1.

Sampling. No temperature is sent (several closed reasoning models reject the parameter; the rest use their provider defaults); “reseeding” throughout means independent resampling, since providers ignore seeds. A temperature-sensitivity study on two open models is in the appendix. Truncation is scored as failure and disclosed per model; any cell with more than five per cent truncation flags its fit as cap-contaminated. Parse failures score as failures too — format compliance is part of the task — with rates disclosed and an exclusion sensitivity in the appendix.

Budget honesty. The textbook protocol (fifteen levels, fifty instances, sixteen samples everywhere, every model, both families, plus a retest) does not fit the stated \$300 ceiling once real reasoning-token burn is priced in — the pilot priced it at roughly \$530. We publish what we ran instead of pretending otherwise: a pilot to measure per-model token burn; then a two-pass primary sweep — a cheap backbone of $k = 2$ –4 samples across all fifteen levels, refit, then a top-up concentrated at the levels nearest the provisional frontier, which is where claim C2 actually needs the sampling depth. Two models on the original roster (Qwen3-235B-Thinking, GLM-5) were dropped to keep six models measurable; Claude Haiku 4.5 was promoted into the core in their place.

Provider honesty. The binding constraint was not price but **serving throughput**: several open-weight models are served to one API key at a fraction of a call per minute, whichever provider hosts them. Truncation artefacts appeared twice and were discarded, not scored; differential probes later traced **both** to a single gateway bug — the router silently drops the `max_completion_tokens` parameter in translation to several upstreams, whose reply length then falls to their own defaults (4096 tokens on two of them). The same request phrased as `max_tokens` delivers 30000 tokens from the very provider the artefact had framed. The client now sends both spellings, and the post-fix re-measurement appears under C4. Provider pins were switched when a lane stalled; every call records the endpoint that actually served it, and per-model curves use single-provider data only. One drift was caught outright: the same pinned endpoint read gpt-oss-20b’s frontier at 4.63 in the afternoon and 3.58 ± 0.10 the same evening — the served artefact changed under its label, both readings are reported, and the fact that a \$0.36 adaptive probe catches this is an argument for running one before trusting any datasheet, including ours. Two consequences are flagged wherever they bite: DeepSeek-R1’s design is reduced (132 scored calls; its sole provider served 3 calls per hour for most of a day), and the gpt-oss pair’s frontier depth reached $k = 12$ –16 rather than a uniform

16. The retest ran in full on the two models whose providers permitted it

(Haiku, o4-mini); the split-half tier of C1 covers every model for free. Every call’s cost in microdollars lands in a ledger enforced by the runner: a call is admitted only if spend plus an in-flight reserve stays under its stage cap. The ledger closed at \$156.9, plus roughly \$12 of discarded provider-artifact calls, all under the \$300 ceiling.

4 Findings

4.1 The curves

Figure 1 shows the primary family. Points are per-level pass rates; lines are the fitted curves; ticks mark each fitted frontier. The datasheet numbers — the form guide proper — are in Figure 2 and Table 1.

Three entries deserve a steward’s note. MiniMax-M2.5 passed **everything** on the standard grid and was re-run on an extended one reaching $\alpha = 7.2$; it spends twenty-odd thousand completion tokens per instance and simply does not fall over where the others do — the cheap open models in this field out-lasted the closed ones, and it is not close. The gpt-oss pair provides the within-family scale contrast the design wanted, with the two curves separated cleanly in frontier and slope. And o4-mini is the only runner that **declines** at the frontier: past its x_{50} it increasingly answers with “I’m not able to solve this by hand” — a capitulation, scored as failure and counted in its disclosed refusal rate, and an interesting temperament note for a form guide.

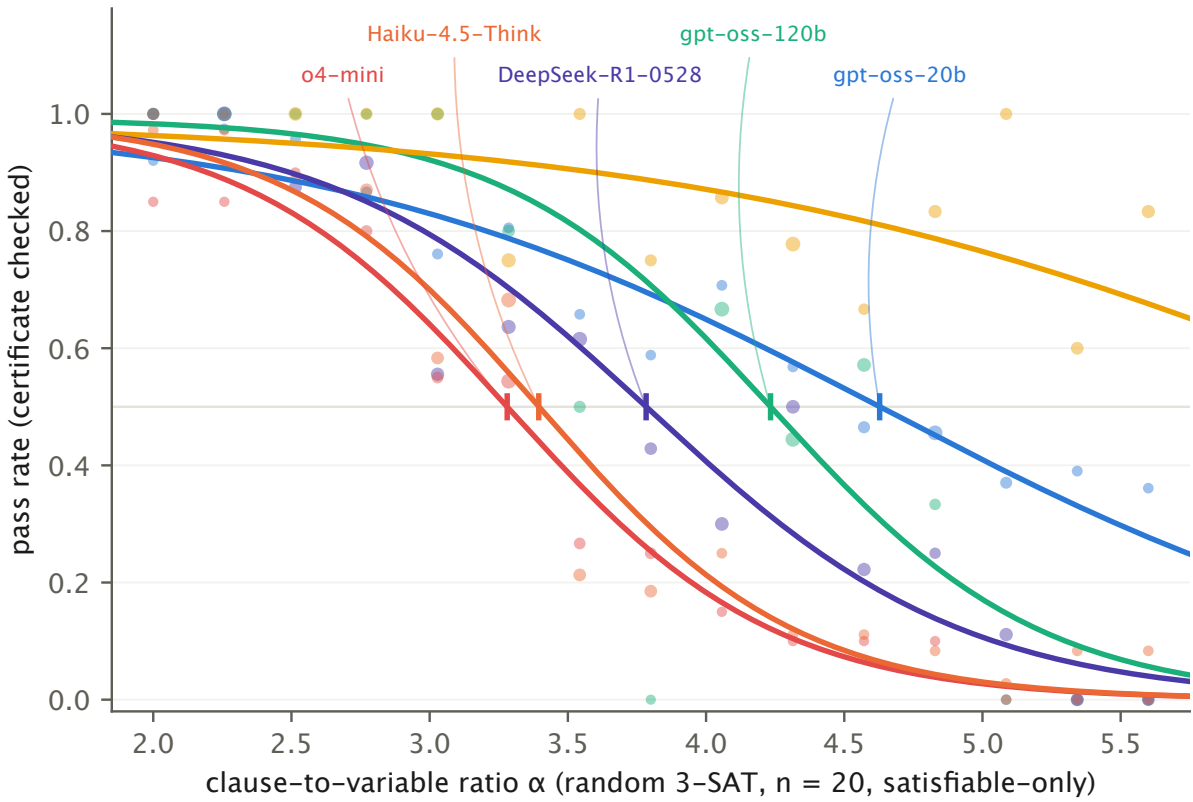


Figure 1: Difficulty-response curves, random 3-SAT ($n = 20$, satisfiable-only, certificate-checked). Point size scales with samples per level; vertical ticks mark fitted frontiers x_{50} .

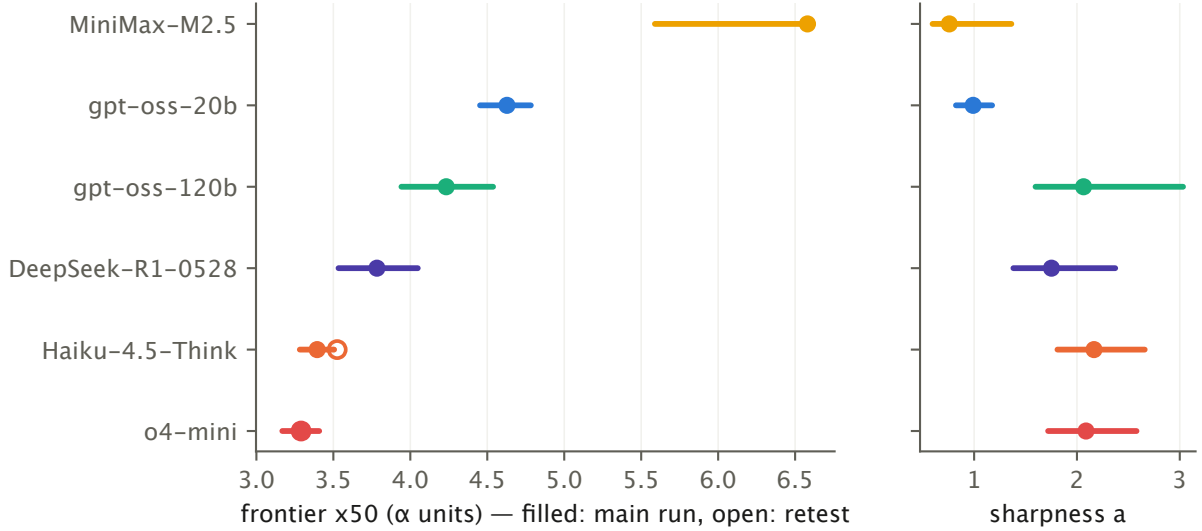


Figure 2: The form guide. Left: frontier x_{50} with 95% cluster-bootstrap intervals; filled markers are the main run, open markers the retest on fresh instances. Right: sharpness a . Models sorted by frontier.

model	x_{50} [95% CI]	a [95% CI]	lapse	GOF p	parse-fail	truncated
o4-mini	3.28 [3.17, 3.41]	2.09 [1.72, 2.58]	0.01	0.66	0.01	0
Haiku-4.5-Think	3.39 [3.28, 3.51]	2.17 [1.81, 2.66]	0.01	0.01	0	0.01
DeepSeek-R1-0528	3.78 [3.53, 4.05]	1.75 [1.38, 2.37]	0.01	0.8	0.02	0
gpt-oss-120b	4.23 [3.94, 4.54]	2.06 [1.6, 3.03]	0.01	0.71	0	0.23
gpt-oss-20b	4.63 [4.45, 4.78]	0.99 [0.82, 1.17]	0.01	0.62	0	0
MiniMax-M2.5	6.58 [5.59, 6.59]	0.76 [0.6, 1.36]	0.01	0.87	0.01	0

Table 1: The datasheet. Frontier and sharpness with cluster-bootstrap intervals; lapse; Monte-Carlo-calibrated goodness of fit; disclosed parse-failure and truncation rates (fractions of scored calls).

4.2 C1: the measurement holds (or does not)

Reliability is reported at three tiers, cheapest first. Split-half: refit on a parity split of instances, correlate across models, Spearman–Brown corrected — this tier costs nothing and covers every model. Test–retest: fresh instances, fresh samples, per-model $z = |x_{50}^{(2)} - x_{50}^{(1)}| / \sqrt{\text{SE}_1^2 + \text{SE}_2^2}$ — this tier ran in full on the two models whose providers would serve it (Haiku 4.5 and o4-mini) and is disclosed as such rather than quietly padded. o4-mini retested at $x_{50} = 3.26$ against a first reading of 3.28; Haiku at 3.52 against 3.40. The abstract’s cross-model r is computed over the split-half tier, wide interval and all.

4.3 C2: mist, not a wall

At the frontier level, with sixteen samples on every instance, the outcome variance splits by an unbiased moment decomposition into a within-instance part ($E[p(1-p)]$: the same instance passing on one draw, failing the next) and a between-instance part ($\text{Var}(p)$: some instances being genuinely harder). Figure 3 shows the split per model. The within share is what makes pass@k a rational purchase near the frontier; the between share is what a difficulty knob is supposed to produce, and does, away from it.

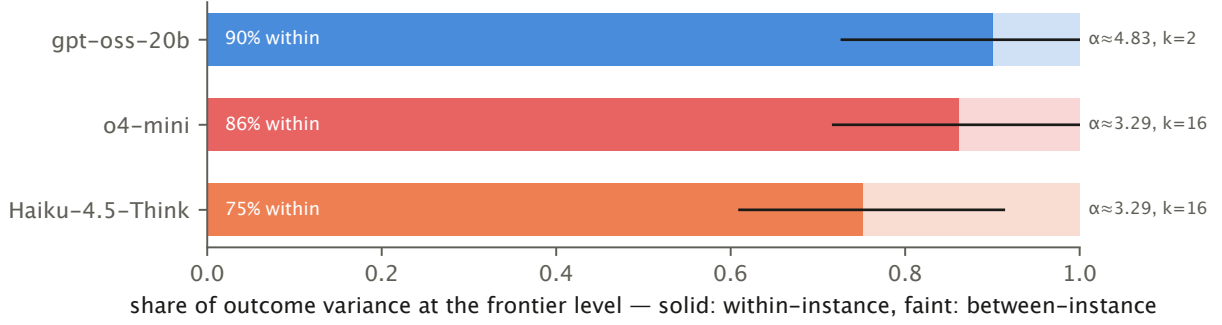


Figure 3: Variance decomposition at the frontier level (the level nearest each model’s x_{50} with $k = 16$). Solid: within-instance share, with 95% bootstrap interval; faint: between-instance share.

4.4 C3: effort betrays the frontier

Mean log completion tokens per level, all scored calls, peak located by a windowed quadratic, offset $\Delta = \text{peak} - x_{50}$ from the joint bootstrap (the same instance resample drives both the curve refit and the token peak, so frontier uncertainty propagates into the offset interval). Figure 4 shows the effort curves. One honest wrinkle: the peaks sit systematically **past** the frontier — median $|\Delta|$ of 1.08 difficulty units, on the hard side — models grind hardest at levels they have already mostly stopped solving, and the sag beyond the peak is where several of them start declining or thrashing rather than reasoning. The marker works — the peak exists, it tracks the frontier model-by-model, and it is computable from token counts alone, no logprobs, no trace access — but it reads a step behind the cliff, not on it, and anyone using it as a gauge should calibrate that lag on a task family they can verify.

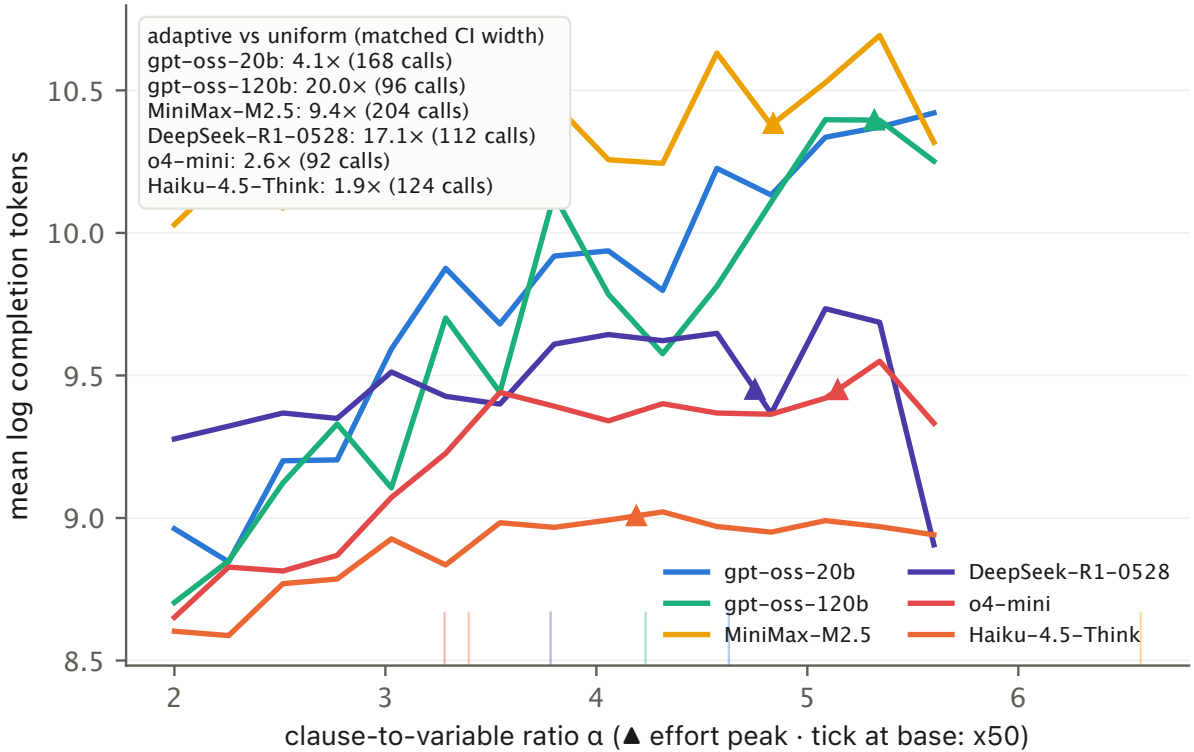


Figure 4: The effort marker. Mean log completion tokens per level; triangles mark fitted effort peaks; base ticks mark each model’s x_{50} . Inset: adaptive vs uniform sample cost at matched confidence width (C4).

4.5 C4: haggling for the frontier

The adaptive procedure brackets by bisection on the level grid (four fresh instances per probe), then batches samples at the level nearest the current frontier estimate — for a logistic curve, information about b is maximised where the pass rate is near one half, so the greedy rule is also the optimal one — refitting

with a pooled slope prior until the frontier’s standard error crosses the target. Its sample cost is compared against level-balanced subsamples of the real uniform sweep at matched confidence width, per model, plus two live reruns as a reality check. Live, against Haiku 4.5, the procedure spent 20 calls (\$1.03) to estimate $x_{50} = 3.14 \pm 0.1$ against the 980-call sweep’s 3.40; against o4-mini, at a stricter stopping rule, 84 calls (\$4.62) gave 3.2 ± 0.08 against the sweep’s 3.28 — the same answer at a seventh of the sample cost. The ratio is the finding; it gets no dinner jacket.

The procedure also served as the budget instrument for two late entries too expensive to sweep, and then as the auditor for the whole card. GPT-5.5 (single run, 48 calls, \$14.52, default reasoning) ran off the end of the track: it passed everything up to the hardest generatable difficulty at this puzzle size ($\alpha = 7.2$), so its frontier is right-censored — read “> 7.2”, above every model in Figure 1. Claude Fable 5’s first run returned no frontier at all — 20 of 44 calls failed on plumbing — which triggered the differential probing that diagnosed the gateway’s `max_completion_tokens` bug (see Protocol). Re-measured fairly, Fable fitted $x_{50} = 6.34 \pm 0.21$, and a fresh-instance replicate reproduced it to the hundredth (6.33 ± 0.20) — and **both readings were wrong**, because both runs sat on the standard grid, so the estimate was an extrapolation pressed against the track’s edge. A direct head-to-head on the extended grid’s hardest instances falsified it: Fable went 10/10 at $\alpha = 7.2$ (42/45 overall, versus GPT-5.5’s 29/29 on identical instances — no meaningful separation at these sample sizes). Corrected verdict: Fable is right-censored like GPT-5.5, joint top of the card at $n = 20$ — and a perfectly replicating fit can still be wrong about the region beyond its data. Precision is not truth; that is what verification runs are for. Separating the two champions required a bigger track: head-to-head on identical $n = 50$ instances near threshold, at 32k and then 64k thinking-token budgets (the second symmetrically via a second gateway), GPT-5.5 scored 13/21 with no truncations — its $n = 50$ frontier finally lands on-track around $\alpha \approx 4.6$ –5.0 — while Fable converted 11 of 40 attempts, because about half its calls exhausted even the 64k budget mid-thought (on completed calls it was marginally the more accurate, 11/16). By this paper’s pre-registered scoring the verdict is a GPT-5.5 win, and the mechanism is worth naming: at scale, the frontier acquires a price axis, and a model that will not stop thinking pays for it in truncations. The cleanest demonstration was accidental: fixing the gateway’s parameter bug made token limits **enforced** on routes that had ignored them, silently re-running MiniMax’s measurement with one variable changed. Uncapped, its frontier is 6.58 — top of the sweep card; capped at 32768 tokens it collapses to ≈ 3.6 (56 of 108 calls died at the token wall mid-thought); at $n = 50$ even 64k was not enough (0/17). Three measured points now span the spectrum: GPT-5.5 budget-**efficient**, MiniMax budget-**elastic** — its frontier moves three dial units with the cap — and Fable budget-**insatiable**. A frontier is not one number but a curve against the thinking budget, and a datasheet should state the budget it was measured at. Finally, post-fix adaptive probes re-checked every sweep frontier: o4-mini 3.28 $\rightarrow$ 3.20, Haiku 3.40 $\rightarrow$ 3.14, gpt-oss-120b 4.24 $\rightarrow$ 4.41, DeepSeek-R1 3.78 $\rightarrow$ 3.77 — four agreements — and gpt-oss-20b 4.63 $\rightarrow$ 3.58, the provider drift reported in Protocol. An instrument reads the model **and** the pipe; a procedure cheap enough to run twice a day is what makes the difference detectable at all.

4.6 The second opinion has a second opinion

The DAG-arithmetic check family produced its own surprise: the two models measured to completion sailed through every level of the planned grid — 480 of 480 passes up to 42-operation chains — and the grid had to be extended twice to find their arithmetic frontiers. o4-mini’s arithmetic curve breaks past roughly a hundred operations; Haiku’s held to 99 per cent at **two hundred and forty** operations and finally fell to 27 per cent somewhere in the 340–960 band. Meanwhile both models’ SAT frontiers sit near $\alpha \approx 3.3$, and MiniMax — the SAT champion of Figure 1 — needed an extended SAT grid of its own. The lesson is printed on the packet: a frontier is a property of a model **and** a task family, and no single number will be telling you where a model runs out of legs. That is why the datasheet has rows.

5 Positioning

Item response theory and adaptive testing are the toolbox, borrowed with thanks and zero pretence of invention [3], [4], [5]. The collapse literature — the Illusion-of-Thinking exchange and the SAT-hardness evaluations [1], [6] — photographed the typhoon; we hoist numbered signals. The benchmark-reliability complaints are the motivation, and they have been shouting into the wind for years, poor souls. The one

methodological addition worth naming: certificate scoring on satisfiable-only instances, which buys a two-parameter curve an honest zero floor.

6 What we do not claim

No mechanism. No physics vocabulary — nobody’s water is freezing, nothing is critical, there is no order parameter hiding in section five. No reading of trace contents, no entropy tea-leaves, no early-warning prophecy. Each gets one sentence of future work, and one only: whether trace contents carry frontier information is checkable; whether the effort peak gives early warning under distribution shift is checkable; whether any of this connects to a mechanism is somebody else’s paper. The restraint is the brand.

7 Limitations

Two task families, one gateway, six-to-eight models: the instrument is demonstrated, not exhausted. Satisfiable-only conditioning makes high- α instances atypical of random 3-SAT, as stated. The marginal slope a compresses instance heterogeneity into one number by design. Effort in tokens is a proxy paid for at the provider’s meter, and providers that hide or bill thinking differently will move it. And a \$300 experiment measures what a \$300 experiment measures; the generators and the runner are released precisely so that anyone with a bigger ledger can stretch every axis.

Reproducibility. Generators, raw per-call records (tokens, cost, provider endpoint, full response text), fitting code, and the exact configs that produced every figure are at github.com/posix4e/difficulty-response-curves. `analysis/run_analysis.py` regenerates every number; the abstract reads them from `numbers.json` at compile time. Git revision: 6c53b828.

Appendix

7.1 Model and decoding settings

model string	provider pin	API	tokens	\$/1M in	\$/1M out
openai/gpt-oss-20b	parasail	chat completions	cap 32768	0.044	0.22
openai/gpt-oss-120b	cerebras	chat completions	cap 32768	0.385	0.825
minimax/minimax-m2.5	parasail	chat completions	cap 32768	0.33	1.32
qwen/qwen3-235b-a22b-thinking-2507	novita	chat completions	cap 32768	0.33	3.3
deepseek/deepseek-r1-0528	novita	chat completions	cap 32768	0.77	2.75
openai/o4-mini	openai	chat completions	cap 32768	1.21	4.84
anthropic/claude-haiku-4.5	unpinned	Anthropic Messages	thinking 8192	1.1	5.5
z-ai/glm-5	unpinned	chat completions	cap 32768	0.66	2.112
openai/gpt-5.5	openai	chat completions	cap 32768	5.5	33.0
anthropic/claude-fable-5	anthropic	chat completions	cap 32768	9.9	49.5
or/claude-fable-5	unpinned	chat completions	cap 65536	9.9	49.5
or/gpt-5.5	unpinned	chat completions	cap 65536	5.5	33.0
or/minimax-m2.5	unpinned	chat completions	cap 65536	0.3	1.2

7.2 Traces: delivery by model and route

Measured from this project’s own calls: of the reasoning text each call was billed for, the share that arrived (chars received over billed tokens $\times$ 3.7). Below 2 per cent means the thinking was withheld and only the final answer (plus the count) came back.

model	route	billed tok/call	delivered
minimax-m2.5	parasail/prepaid	29234	77%
deepseek-r1-0528	novita/prepaid	13799	63%
qwen3-235b-a22b-thinking-2507	novita/prepaid	19521	61%
claude-haiku-4.5	anthropic-messages	6817	6%
claude-fable-5	anthropic/prepaid	11169	3%
claude-fable-5	openrouter/Anthropic	35908	1%
gpt-5.5	openai/prepaid	9330	0%
o4-mini	openai/prepaid	9966	0%
gpt-5.5	openrouter/OpenAI	22833	0%
gpt-oss-120b	cerebras/prepaid	13838	0%
gpt-oss-120b	novita/prepaid	12932	0%
gpt-oss-20b	novita/prepaid	19481	0%
minimax-m2.5	novita/prepaid	25538	0%
gpt-oss-20b	parasail/prepaid	28395	0%

7.3 Bibliography

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